

NAME:PRENA .M

REG NO:192524056

COURSE NAME:JAVA PROGRAMMING

COURSE CODE:CSA 0924

CO3-AT3

Differentiate daemon and user threads. A background cache-refresh thread prevents JVM shutdown. Diagnose the problem and fix it.

AIM:

To demonstrate the difference between **daemon and user threads** in Java and fix the JVM shutdown problem caused by a background cache-refresh thread by setting it as a **daemon thread**.

PSEUDOCODE:

START

CREATE a background cache-refresh thread

SET the thread as DAEMON

START the daemon thread

DISPLAY "Main user thread started"

PERFORM main thread task

WAIT for some time

DISPLAY "Main user thread finished"

STOP

END

JAVA CODE:

```
class DaemonThreadDemo {  
    public static void main(String[] args) throws InterruptedException {  
        Thread cacheRefresh = new Thread(() -> {  
            while (true) {  
                System.out.println("Cache refresh running...");  
                try {  
                    Thread.sleep(1000);  
                } catch (InterruptedException e) {  
                    break;  
                }  
            }  
        });  
  
        // Make background thread a daemon  
        cacheRefresh.setDaemon(true);  
        cacheRefresh.start();  
  
        // User thread  
        System.out.println("Main user thread started.");  
        Thread.sleep(3000);  
        System.out.println("Main user thread finished.");  
    }  
}
```

INPUT:

No user input required.

OUTPUT:

Main user thread started.

Cache refresh running...

Cache refresh running...

Cache refresh running...

Main user thread finished.

DaemonThreadDemo.j...	Run	Output
<pre>1 class DaemonThreadDemo { 2 3 public static void main(String[] args) throws InterruptedException { 4 5 Thread cacheRefresh = new Thread(() -> { 6 while (true) { 7 System.out.println("Cache refresh running..."); 8 try { 9 Thread.sleep(1000); 10 } catch (InterruptedException e) { 11 break; 12 } 13 } 14 }); 15 16 // Make background thread a daemon 17 cacheRefresh.setDaemon(true); 18 cacheRefresh.start(); 19 20 // User thread 21 System.out.println("Main user thread started.");</pre>		<pre>Main user thread started. Cache refresh running... Cache refresh running... Cache refresh running... Cache refresh running... Main user thread finished. === Code Execution Successful ===</pre>



DaemonThreadDemo.j...



Share

Run

Output

Clear

```
4      Thread cacheRefresh = new Thread(() -> {
5~
6~          while (true) {
7~              System.out.println("Cache refresh
8~                  running...");
9~              try {
10~                  Thread.sleep(1000);
11~              } catch (InterruptedException e) {
12~                  break;
13~              }
14~          });
15
16      // Make background thread a daemon
17      cacheRefresh.setDaemon(true);
18      cacheRefresh.start();
19
20      // User thread
21      System.out.println("Main user thread started."
22          );
23      Thread.sleep(3000);
24
25      System.out.println("Main user thread finished
26          .");
27  }
```

```
Main user thread started.
Cache refresh running...
Cache refresh running...
Cache refresh running...
Main user thread finished.

=== Code Execution Successful ===
```